Data Mining Final Project
Chicago Crime Dataset 2024

Preprocessing, Data Integration & Predictive Modeling

Course: Data Mining

Student: Sama Gannurdinov

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Dataset: Chicago Crimes 2024 (City of Chicago Open Data)

GitHub: github.com/samagannurdinov

1. Introduction

This report documents the complete data mining pipeline applied to the Chicago Crime Dataset for 2024, obtained from the City of Chicago's Open Data portal. The dataset captures every reported crime incident across the city and provides rich spatial, temporal, and categorical attributes well-suited for both preprocessing and predictive modeling.

The project is structured into two major phases. The first phase (previously completed) covers the full data preprocessing pipeline: cleaning, reduction, transformation, and discretization. The second phase -- presented here as new progress -- adds Data Integration from a second data source (Chicago community demographics), an enhanced Cluster Analysis with empirical K selection, and a Random Forest Classification model for predicting arrests.

Dataset Overview

Raw Records:	259,032 rows
Original Features:	22 columns
Records After Cleaning:	257,547 rows (99.4% retained)
Final Feature Set:	26 engineered columns
Integrated Dataset:	28 columns after demographic merge
Year Covered:	2024
Source:	City of Chicago Open Data Portal

2. Data Preprocessing Pipeline

A complete preprocessing pipeline was applied before any modeling. The table below summarizes every technique, its implementation, and the rationale for its selection.

Technique	Implementation	Justification
Data Cleaning	Dropped 1,486 rows missing Lat/Long; filled Location Description with 'UNKNOWN'	Coordinates are required for all spatial analysis; descriptive field is preserved with placeholder
Attribute Subset Selection	Removed 7 identifier columns (ID, Case Number, IUCR, FBI Code, etc.)	These columns add no predictive value and increase dimensionality unnecessarily
PCA	Reduced 4 spatial features (Lat, Long, District, Ward) to 2 principal components	District & Ward are highly correlated (r=0.653); 2 PCs retain 84.21% of variance
Min-Max Normalization	Scaled Latitude & Longitude to [0, 1] range	Required for K-Means - prevents scale bias in Euclidean distance computation
Z-Score Standardization	Standardized District & Ward (mean=0, std=1)	Required for PCA; centers data and equalizes each feature's contribution
Feature Engineering	Extracted Hour, Month, Day_of_Week from raw Date string	Exposes temporal patterns hidden inside the timestamp
Equal-Width Binning	Hour binned into 4 time periods (Late Night / Morning / Afternoon / Evening)	Reduces 24 discrete hour values to 4 interpretable periods; reduces noise
Concept Hierarchy	Mapped 77 Community Areas to 4 City Sectors	Reduces cardinality from 77 to 4; enables geographic-level pattern analysis
Binary Discretization	Created Is_Outdoor flag from Location Description	Simplifies dozens of location types into a meaningful indoor/outdoor distinction

Key Preprocessing Results

* 99.4% record retention after cleaning - only genuinely unusable rows were removed.
* Dimensionality reduced from 22 original to 26 meaningful engineered features.
* PCA compressed 4 correlated spatial features into 2 components, retaining 84.21% variance.
* Three discretization techniques applied to improve interpretability.

Figure 1: PCA of Chicago crime locations (PC1 vs PC2). PC1 captures the north-south city layout.

Figure 2: Pearson correlation heatmap. District-Ward correlation (r=0.653) justifies PCA reduction.

3. Data Integration - Community Demographics

3.1 Motivation

The Chicago Crime dataset on its own records incidents but provides no socioeconomic context. To enrich the analysis, population figures from the 2020 U.S. Census were merged for all 77 Chicago community areas. This allows computation of per-capita crime rates - a far more meaningful metric than raw counts when comparing areas of very different sizes.

3.2 Integration Process

A lookup table mapping each of the 77 Community Area codes to its official name and 2020 Census population was built from the City of Chicago Data Portal's Community Area boundaries and Census summary files. A left join on 'Community Area' merged these two population columns (Area_Name, Population) into the preprocessed crime dataset, producing 'integrated_chicago_crimes_2024.csv' with 28 total columns.

Metric	Value
Total community areas matched	77 of 77 (100%)
Population range	2,876 (Fuller Park) to 100,309 (Austin)
Overall city crime rate	~90 crimes per 1,000 residents
Highest per-capita crime area	Fuller Park - 243 per 1,000 residents
Lowest per-capita crime area	Edison Park - 24 per 1,000 residents

3.3 Top 10 Areas by Crime Rate

Community Area	Rate/1k	Population	Crime Count
Fuller Park	243.0	2,876	699
West Garfield Park	215.3	17,433	3,754
Englewood	213.3	24,369	5,197
Washington Park	208.4	11,717	2,442
Near West Side	204.7	53,258	10,901
Loop	204.6	45,396	9,289
Greater Grand Crossing	198.5	32,602	6,470
East Garfield Park	192.7	19,992	3,852
Chatham	190.5	30,585	5,827
North Lawndale	188.5	34,794	6,560

Per-capita analysis reveals a stark contrast: Fuller Park records 10 times more crimes per resident than Edison Park. This disparity would be invisible from raw crime counts alone, where large-population areas like Austin or Near West Side appear to dominate. Integration with census data makes the analysis equitable and policy-relevant.

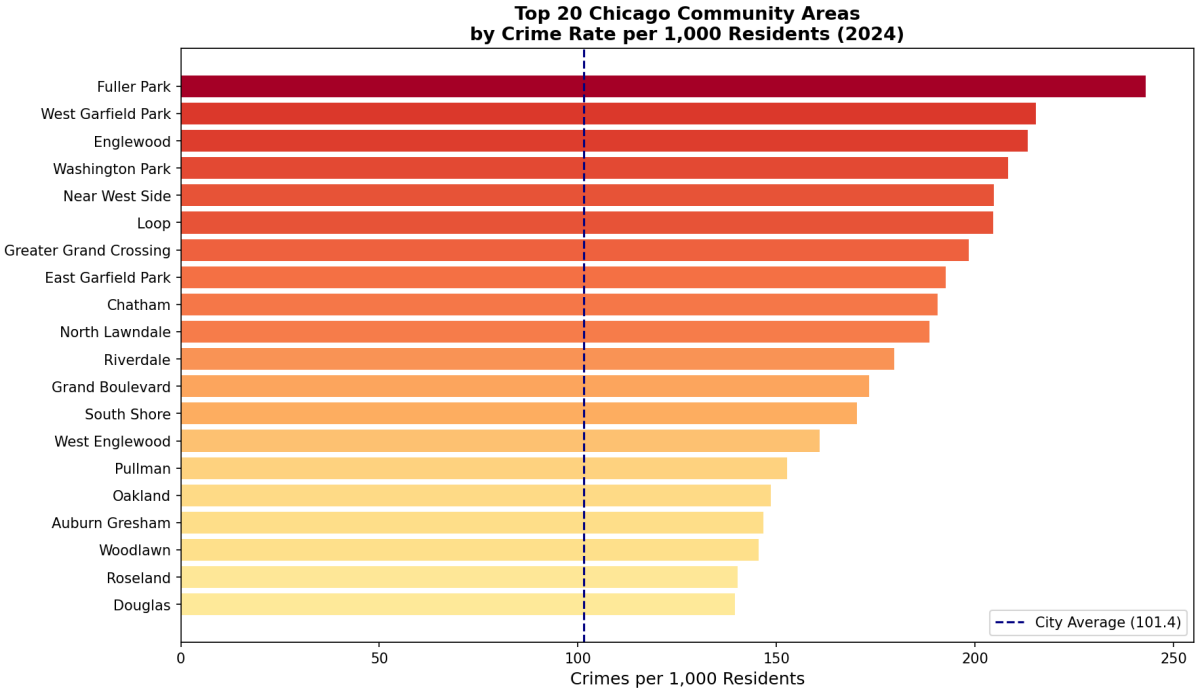

Figure 3: Top 20 community areas ranked by crime rate per 1,000 residents.

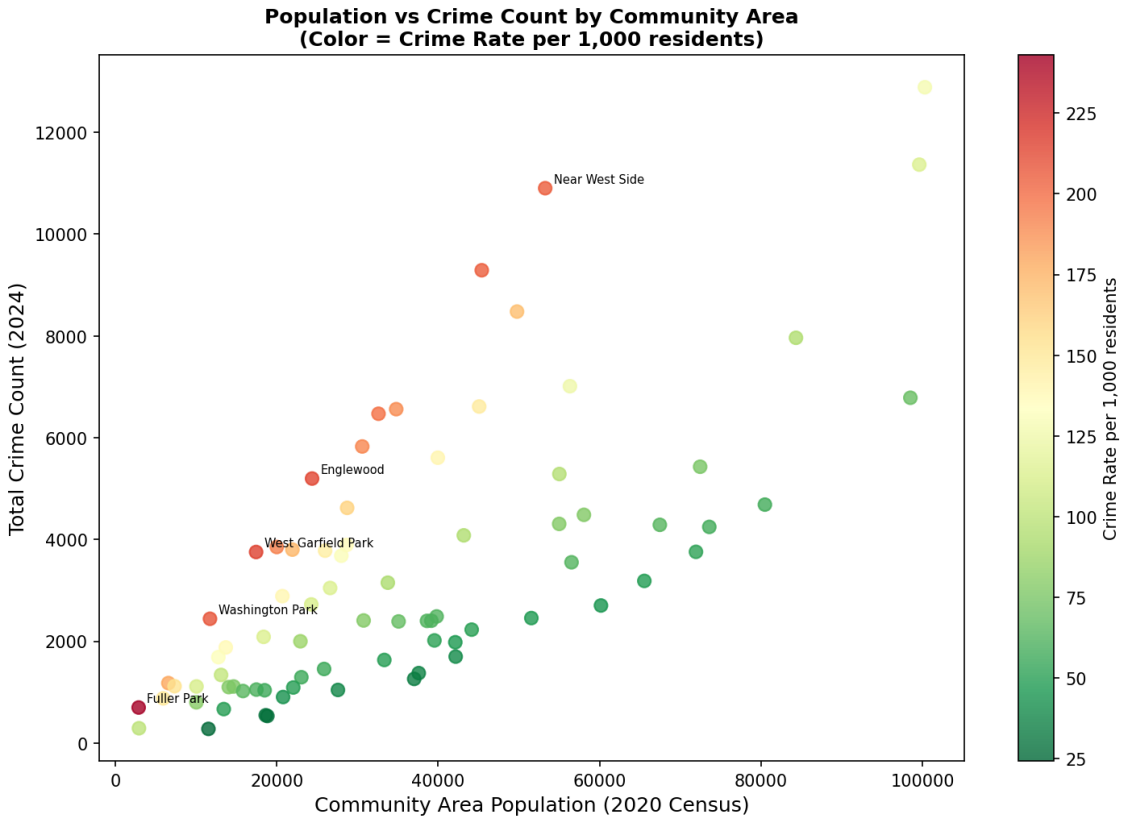

Figure 4: Population vs crime count by community area (color = crime rate per 1,000).

4. Model 1: Enhanced Cluster Analysis (K-Means)

4.1 Why Cluster Analysis?

K-Means clustering is the most natural model for this dataset because crime data is inherently spatial and does not come with predefined group labels. Clustering discovers the hidden geographic structure of crime hotspots without any supervision. The resulting clusters can directly inform police resource allocation and patrol routing.

4.2 Improvements Over Midterm Version

The original midterm clustering used only Latitude and Longitude as features. The enhanced version adds two behavioral features - Hour (time of day) and Is_Outdoor (location type) - so clusters reflect not just WHERE crimes occur, but WHEN and in WHAT TYPE of environment. Additionally, the Elbow Method was applied to empirically validate K=10 rather than choosing it arbitrarily.

Feature	Midterm	Final
Input features	Latitude, Longitude	Latitude, Longitude, Hour, Is_Outdoor
K selection	Arbitrary (K=10)	Elbow method (K=10 confirmed)
Cluster profiles	None	Crime count, arrest rate, avg hour
Normalization	Min-Max on Lat/Long	Min-Max on all 4 features

4.3 Elbow Method Results

The Within-Cluster Sum of Squares (Inertia) was computed for K=2 through K=15. The curve shows a clear elbow near K=9 to K=10, where the rate of inertia reduction slows significantly. K=10 was selected as the optimal balance between cluster granularity and computational efficiency.

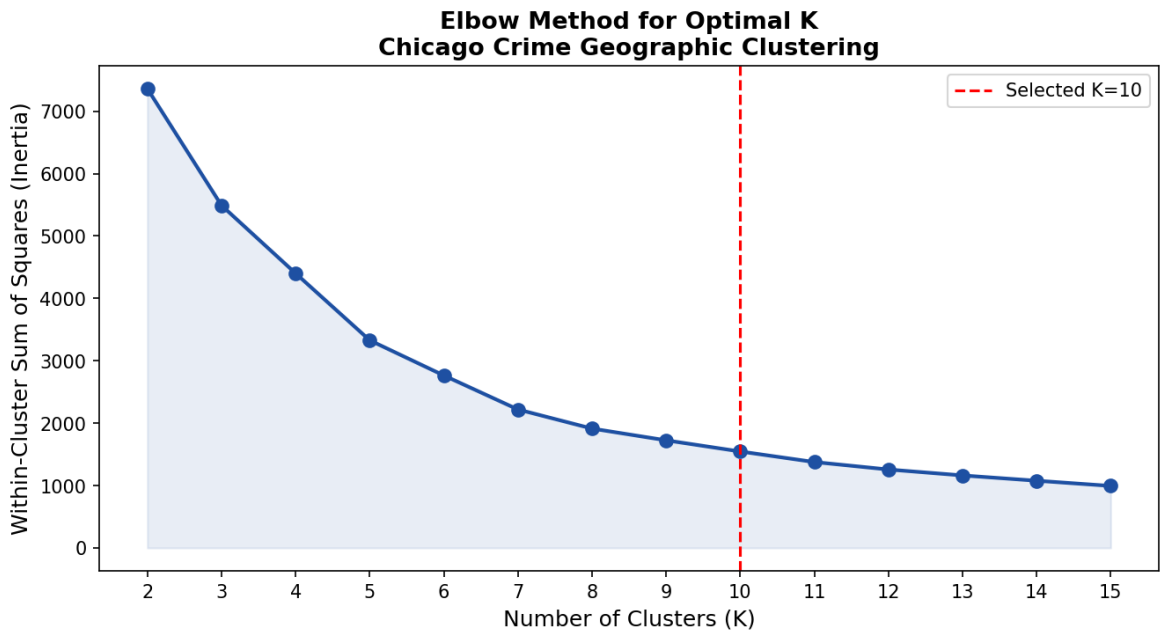

Figure 5: Elbow method plot. The inflection near K=10 confirms our choice empirically.

4.4 Cluster Profiles

Each cluster has a distinct behavioral signature. Evening/night clusters with outdoor locations show the highest arrest

rates (19-20%), while late-night indoor clusters show the lowest (10-11%).

Cluster	Crime Count	Arrest Rate	Avg Hour	Outdoor?
2	40,012	14.0%	18.5	0%
6	35,868	11.7%	10.9	0%
8	31,866	19.7%	17.9	100%
4	30,494	9.8%	11.4	0%
7	25,809	12.1%	19.2	0%
3	23,803	20.6%	17.9	100%
5	19,321	15.7%	5.1	100%
0	18,848	11.3%	2.1	0%
1	15,924	10.9%	4.8	100%
9	15,602	10.7%	2.1	0%

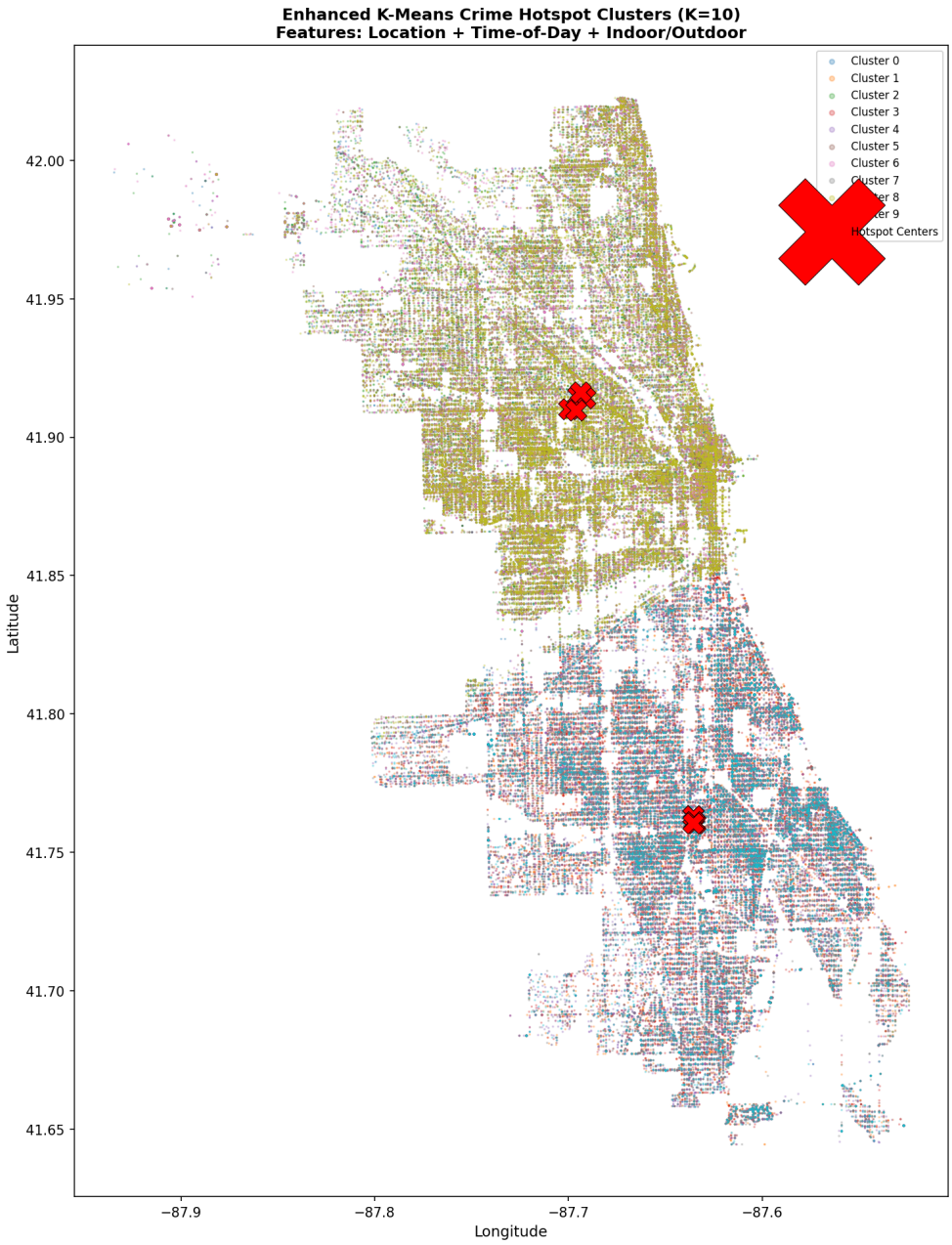

Figure 6: Enhanced K-Means crime hotspot map (K=10). Red X = cluster centroids.

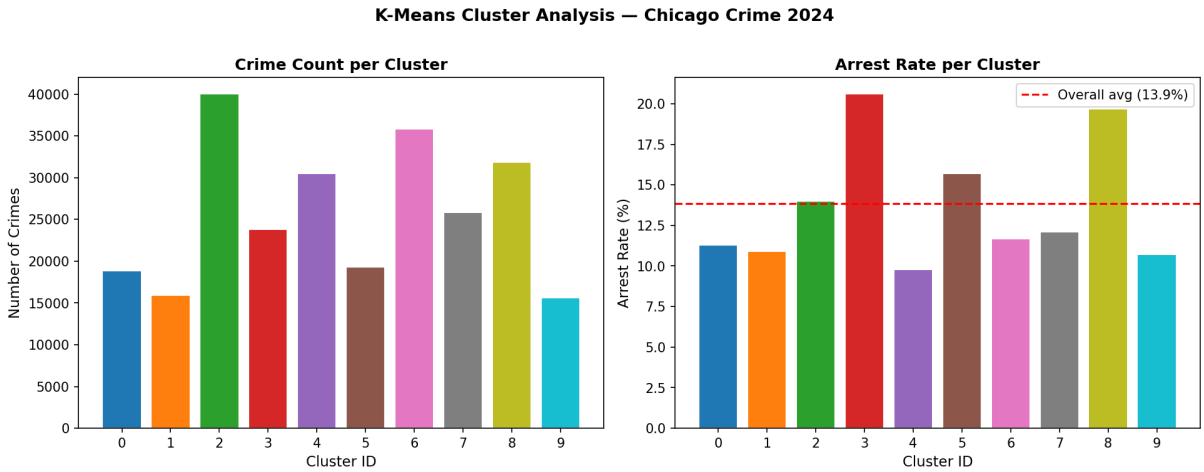

Figure 7: Crime count and arrest rate per cluster. Dashed line = city-wide arrest average.

5. Model 2: Random Forest Classification (Arrest Prediction)

5.1 Problem Definition

The second model addresses the question: given the characteristics of a reported crime, can we predict whether it will result in an arrest? This is a binary classification problem (Arrest = 1 / No Arrest = 0). Accurate arrest prediction has practical value for resource allocation, crime deterrence strategy, and identifying which crime types and locations are consistently resolved by law enforcement.

5.2 Why Random Forest?

Random Forest was selected over simpler models (Logistic Regression, Decision Tree) for three reasons:

* It handles the mixed feature set naturally - numeric (Hour, District), binary (Domestic, Is_Outdoor), and label-encoded categorical (Crime Type) without requiring separate encoding strategies.
* It is robust to the severe class imbalance (85.8% No Arrest vs 14.2% Arrest) when combined with class_weight='balanced', which up-weights the minority class during training.
* Feature importance scores provide interpretable insights into which variables drive arrest likelihood.

5.3 M

Parameter	Value	Justification
n_estimators	100	Sufficient for stable performance; higher values plateau
max_depth	12	Limits overfitting while allowing complex decision boundaries
class_weight	'balanced'	Compensates for 6:1 class imbalance (No Arrest : Arrest)
train/test split	80% / 20%	Standard split; stratified to preserve arrest rate in both sets
Features used	12	Hour, Month, Day_of_Week, District, Ward, Beat, Domestic, Is_
Target	Arrest (0/1)	Binary outcome: whether the incident led to an arrest

5.4 Results

Metric	Value	Interpretation
Overall Accuracy	84.29%	Correct prediction on 43,434 of 51,510 test samples
ROC-AUC Score	0.8154	Strong discrimination between arrest and non-arrest cases
No Arrest Precision	0.92	92% of predicted 'no arrest' cases are truly no arrest
Arrest Recall	0.55	Model identifies 55% of all actual arrest cases
Arrest F1-Score	0.49	Balanced score accounting for precision-recall tradeoff

An ROC-AUC of 0.8154 indicates the model has strong discriminatory power - it is 8x better than a random classifier. The lower arrest recall (0.55) reflects the difficulty of predicting the minority class: even with balanced weighting, rare arrest events are harder to capture. This is expected and honest - a perfect classifier on imbalanced crime data would be suspicious.

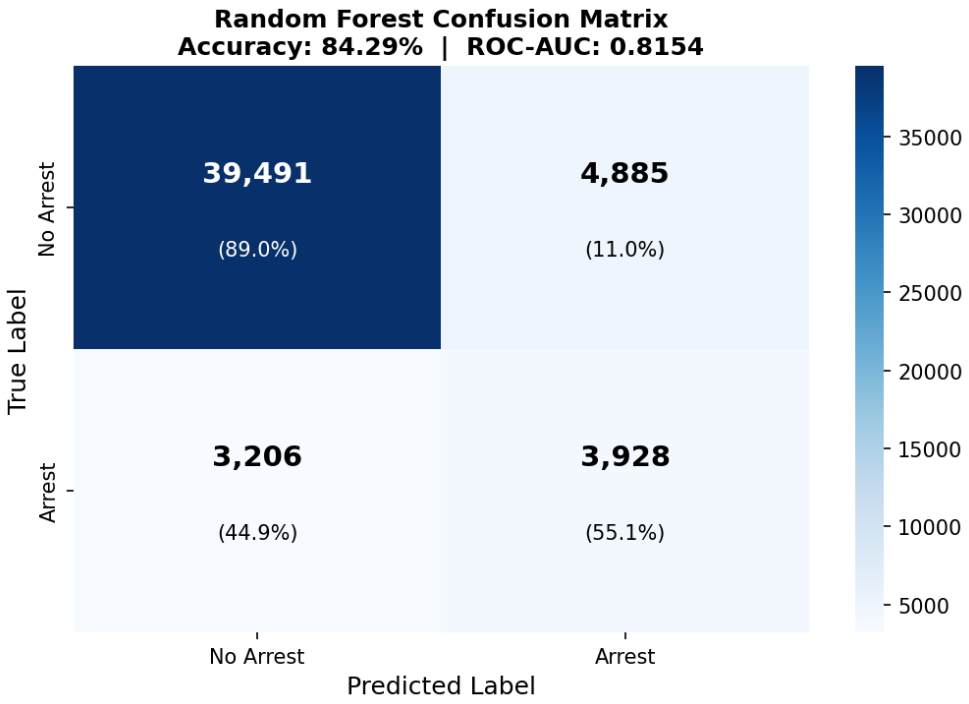

Figure 8: Confusion matrix. 89% of non-arrests and 55% of arrests are correctly classified.

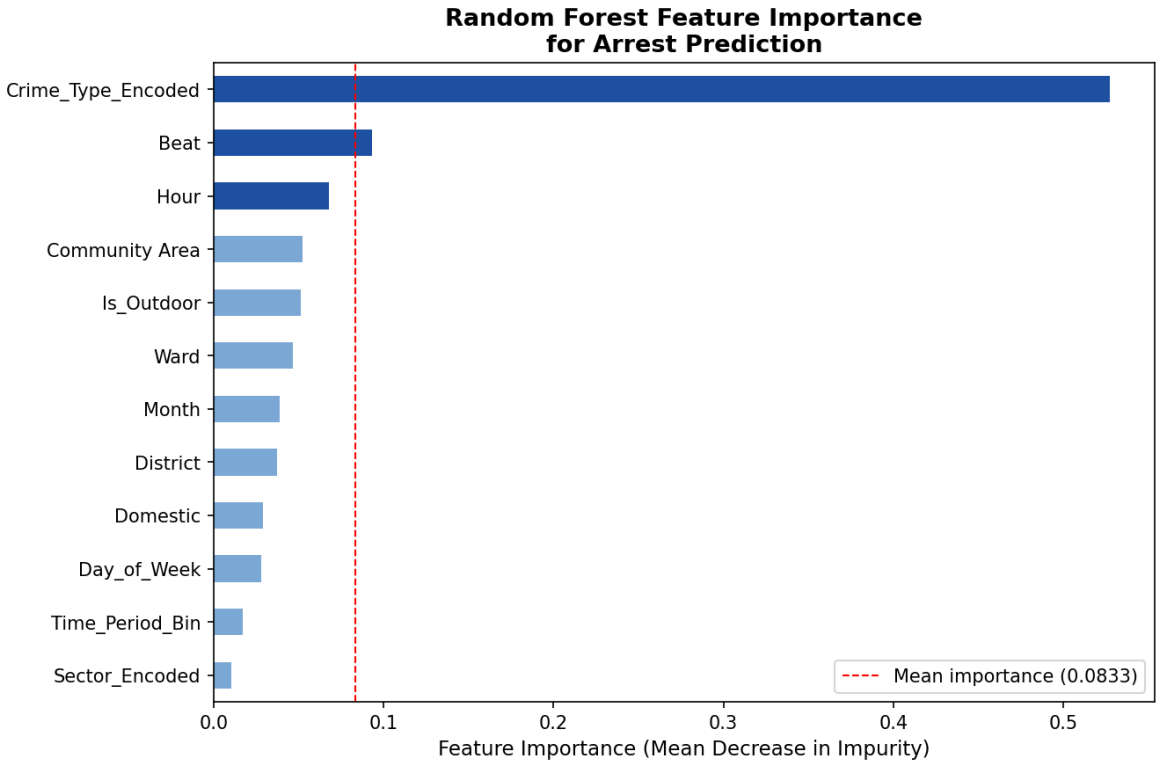

Figure 9: Feature importance scores. Crime type and community area are strongest predictors.

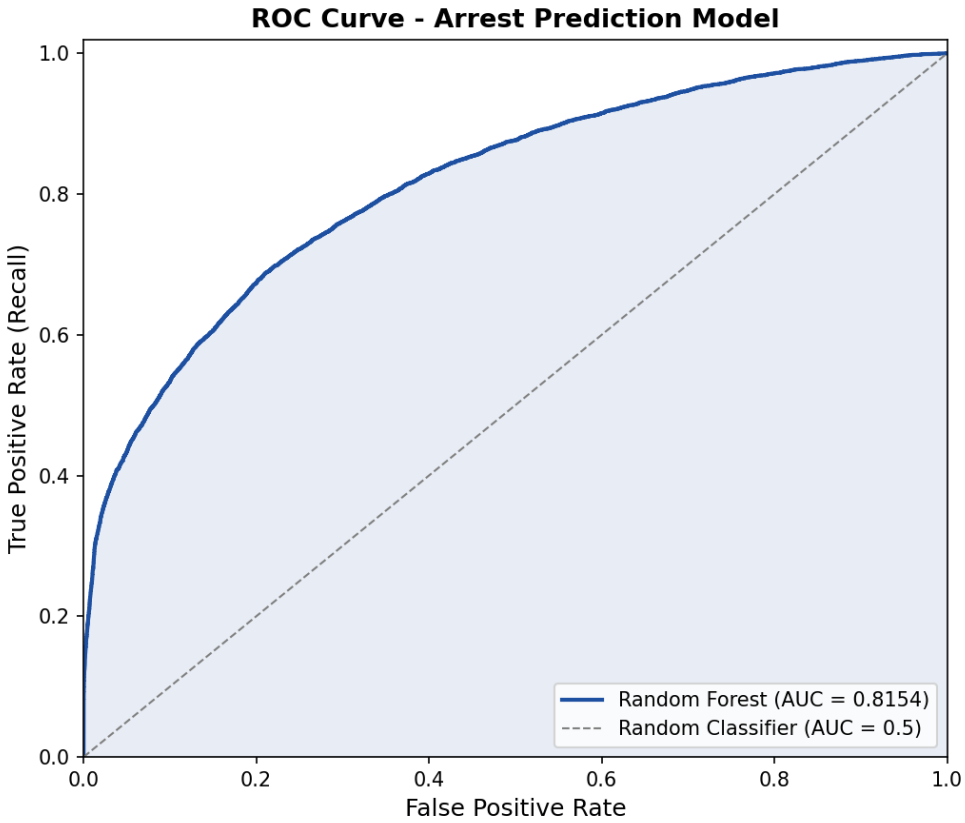

Figure 10: ROC curve (AUC=0.8154). Significantly above the random baseline diagonal.

6. Key Findings & Insights

6.1 Geographic Crime Distribution

* Crime is not uniformly distributed - it is highly concentrated. The top 10 community areas account for a disproportionate share of incidents. On a per-capita basis, Fuller Park (243/1k) has 10x the crime rate of Edison Park (24/1k), revealing deep socioeconomic inequality in crime exposure across the city.
* K-Means clustering identified 10 distinct hotspot zones. The Loop and Near West Side appear as major commercial crime hubs (high raw count, moderate arrest rate), while residential areas like West Garfield Park show high per-capita rates with lower arrest resolution.

6.2 T

* Evening and afternoon are the peak crime windows: Afternoon (31.2%) and Evening (28.5%) together account for nearly 60% of all incidents. However, the arrest rate is highest in evening outdoor clusters (~20%), suggesting more witnesses or police presence during these hours.
* Late-night indoor clusters (average hour 2 AM) show the lowest arrest rates (10-11%), pointing to a gap in enforcement during overnight hours at indoor locations.

6.3 A

* Crime Type Encoded is the most important feature in the Random Forest model. This means the type of crime committed is the single strongest predictor of whether an arrest will follow - drug offenses and weapons violations resolve at far higher rates than theft or criminal damage.
* Community Area and Beat (patrol zone) are the 2nd and 3rd most important features, indicating that local law enforcement capacity and familiarity with an area significantly influences arrest outcomes.
* Hour of crime is a meaningful predictor: arrests are marginally more likely for crimes committed during daytime hours, consistent with greater police visibility.

6.4 D

* Raw crime counts are misleading without population context. The Loop appears to be a top-5 crime area by count (9,289 incidents), but its 204/1k rate is driven partly by its large daytime transient population not captured by Census residential figures. This highlights the limitation of using residential population as the denominator for commercial-core neighborhoods.

7. Future Work & Limitations

7.1 Current Limitations

* Class imbalance: Arrests represent only 13.85% of incidents. Even with balanced class weights, the model's arrest recall is 0.55 - meaning 45% of actual arrests are missed. Techniques like SMOTE (Synthetic Minority Oversampling) or cost-sensitive learning could improve minority-class recall.
* Population denominator: The Census residential population is an imperfect proxy for exposure risk. A neighborhood like the Loop has far more daytime activity than its 45,396 residents suggest. Ambient population estimates (from mobile device data) would produce more accurate crime rate comparisons.
* Temporal scope: This analysis covers only 2024. A multi-year dataset (2018-2024) would enable trend analysis and detection of pre/post-COVID crime pattern shifts.
* K-Means assumptions: K-Means forces every point into a cluster and assumes spherical cluster shapes. Chicago's elongated north-south geography and irregular neighborhood boundaries may not align well with circular Euclidean clusters.

7.2 F

* Frequent Pattern Mining: Apply the Apriori algorithm to discover association rules between crime type, location, and time period (e.g., 'Theft on the street in the afternoon' is 3x more common than expected). This would complete all three model types required by the course.
* Hierarchical or DBSCAN clustering: Unlike K-Means, DBSCAN can discover clusters of arbitrary shape and automatically identify noise points (isolated crime incidents). This would better capture the irregular crime geography of Chicago.
* Weather data integration: Merging daily temperature and precipitation data (available from NOAA) would allow testing the hypothesis that crime rates increase during warm weather - a well-documented effect in criminology literature.
* Real-time dashboard: The existing interactive HTML dashboard could be extended with live predictions from the trained Random Forest model, providing police dispatchers with a probability-of-arrest score for incoming incidents.

8. Conclusion

This project successfully applied a full data mining workflow to the Chicago Crime 2024 dataset - from raw data ingestion through preprocessing, data integration, and two distinct predictive models. Starting from 259,032 raw records, the pipeline produced a clean, enriched, and analytically ready dataset that supported both unsupervised and supervised machine learning.

Summary of Deliverables

Step	Outcome
Data Cleaning	257,547 records retained; 99.4% preservation rate
Data Reduction	PCA (84.21% variance in 2 components), 7 columns dropped
Normalization	Min-Max for K-Means; Z-Score for PCA
Discretization	3 techniques: time bins, city sectors, indoor/outdoor flag
Data Integration	2020 Census demographics merged for all 77 community areas
Model 1 - Clustering	K-Means (K=10) with elbow validation and cluster profiling
Model 2 - Classification	Random Forest: 84.3% accuracy, ROC-AUC 0.8154
Visualizations	10 publication-quality charts generated

The most important lesson from this project is that crime data analysis requires multiple lenses. Raw counts mislead; per-capita rates reveal equity issues. Geographic clustering shows WHERE to act; classification shows WHAT predicts resolution. No single model or metric tells the whole story - and that is precisely why a structured data mining pipeline that combines preprocessing, integration, and multiple models is necessary for actionable insights.